

Two-Observable Decomposition in SU(2) Lattice Gauge Theory: the Cl(0, 2) $\cong \mathbb{H}$ Framework

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Abstract

We present a two-observable decomposition of the SU(2) temporal gauge-link correlator within the Cl(0, 2) $\cong \mathbb{H}$ algebraic framework. The SU(2) parametrisation $U = a \cdot I_2 + b \cdot J$, with $a = (U)_{00} \in \mathbb{C}$ and $b = (U)_{01} \in \mathbb{C}\hat{j}$ (a right $\mathbb{C}$ -module), induces a natural channel decomposition of the full correlator $G_{\text{full}}(t)$ into two bilinears: $G_{\text{diag}}(t) = 8 \sum \text{Re}[a_t \bar{a}_0]$, probing the diagonal (T_3 eigenspace) sector, and $G_{\text{cross}}(t) = 8 \sum \text{Re}[b_t \bar{b}_0]$, probing the off-diagonal sector. This identification resolves a structural gap of Paper 4 of this series: the cross term Δ_{cross} , previously defined subtractionally, is exactly the b -channel bilinear.

Theorem 3.1 (Sec. 3.3) establishes $G_{\text{diag}}(t) + G_{\text{cross}}(t) = G_{\text{full}}(t)$ exactly for every SU(2) field configuration and every timeslice t . Corollary 6.1 (Two-Observable Decomposition, Sec. 6.2) assembles this with the channel faithfulness results — a -channel (Proposition 1 of Paper 4) and b -channel (Proposition 3.1, Sec. 3.1) — into a complete statement of the exact algebraic decomposition. Proposition 5.1 (Sec. 5.2) proves that the rank-1 collapse formula $\Pi_r U \Pi_r = (U)_{rr} \Pi_r$, which underlies the channel separation, is intrinsic to the fundamental SU(2) representation and holds for every tensor-product embedding of U into $\mathbb{C}^8$.

In the weak-coupling expansion, G_{diag} probes the neutral gauge boson field A^3 and G_{cross} probes the charged gauge boson fields $W^\pm$. Proposition 7.1 (Sec. 7.2) establishes G_{cross} as a necessary algebraic conduit for all charged-current coupling. Proposition 8.1 (Sec. 8.1) proves G_{cross} is not gauge-invariant; under Landau gauge fixing,

$$\Delta_{\text{cross}}(t)|_{\text{Landau}}/8V \approx +(g^2/2) C_{W^\pm}(t) \quad \text{at } O(g^2),$$

providing a formal connection to the zero-spatial-momentum $W^\pm$ temporal correlator.

All results are verified by 51 numerical checks (D1–D18), each passing at tolerance $\leq 10^{-13}$ or better. The framework does not distinguish SU(2)_L from SU(2)_R, which requires external fermionic input, and makes no claims about chirality selection, W/Z mass ratios, or the Higgs mechanism.

1 Introduction

1.1 Context

The present work continues a programme of embedding lattice SU(2) gauge theory within the division algebra hierarchy $\mathbb{R} \otimes \mathbb{C} \otimes \mathbb{H} \otimes \mathbb{O}$, exploiting the Clifford algebra isomorphism $\text{Cl}(0, 2) \cong \mathbb{H}$ as the natural algebraic setting for the electroweak SU(2) sector [[P1], [P2], [P4]]. Paper 4 [[P4]] introduced the T_3 -eigenspace projectors $P_{\pm}^{\text{LW}}$ and established two central results for the SU(2) temporal gauge-link correlator. Proposition 1 of Paper 4 [[P4]], §4.2, proved that the diagonal bilinear collapses to a rank-1 form:

$$\text{Re Tr} \left[P_r^{\text{LW}} E_t P_r^{\text{LW}} E_0^\dagger \right] = 4 \text{Re}[a_t \bar{a}_0],$$

where $a_t = (U_t)_{00} \in \mathbb{C}$ is the top-left matrix element of the SU(2) link variable U_t . Corollary 1 of Paper 4 [[P4]] then established that $G_+(t) = G_-(t)$ identically for every $U \in \text{SU}(2)$, so the two eigenspace correlators carry identical information and combine to give the diagonal correlator $G_{\text{diag}}(t)$.

Paper 4 further decomposed the full correlator as $G_{\text{full}}(t) = G_{\text{diag}}(t) + \Delta_{\text{cross}}(t)$, with the cross term Δ_{cross} defined in Paper 4 [[P4]], §4.3, as a residual: $\Delta_{\text{cross}}(t) = G_{\text{full}}(t) - G_{\text{diag}}(t)$. This compact identity was verified numerically across four thermalized ensembles ($\beta_{\text{eff}} \in \{5.0, 6.0, 9.0, 12.0\}$, $N = 6, 20$ configurations) to within $|G_{\text{diag}} + \Delta_{\text{cross}} - G_{\text{full}}| \leq 8.9 \times 10^{-16}$ [[P4]]. The ratio $\Delta_{\text{cross}}/G_{\text{full}}$ was observed to lie in the range $[0.72, 2.33]$ across ensembles—a striking $O(1)$ contribution—and G_{cross} was found to be negative in 37–62% of (configuration, t) pairs [[P4]].

Despite this quantitative success, a structural gap remained. The definition of Δ_{cross} in Paper 4 was purely subtractive: it named a quantity without providing an independent algebraic generator. The question left open was: what is G_{cross} , exactly, in terms of the fundamental SU(2) matrix elements? Answering this question is the task of the present paper.

What is standard and what is new. The parametrisation $U = a \cdot I_2 + b \cdot J$ with $|a|^2 + |b|^2 = 1$ is the standard quaternionic form of an SU(2) matrix; the split of $\text{Tr}[E_t E_0^\dagger]$ into a - and b -channel bilinears follows immediately from expanding the matrix product and is not claimed as new here. What is new in this paper is threefold: (i) the *projector formulation* — identifying the eigenspace projectors $P_{\pm}^{\text{LW}}$ as the algebraic objects that select the a - and b -channels within the $\text{Cl}(0, 2) \cong \mathbb{H}$ framework; (ii) the *Paper 4 gap closure* — proving that the subtractively defined Δ_{cross} equals exactly the b -channel bilinear $8 \sum \text{Re}[b_t \bar{b}_0]$, supplying it with an independent algebraic generator for the first time; and (iii) the *Landau-gauge propagator identification* — the formal $O(g^2)$ connection between G_{cross} and the zero-spatial-momentum $W^\pm$ temporal correlator.

1.2 This Paper

The answer is provided by the off-diagonal rank-1 collapse. Writing the SU(2) link variable in the standard parametrisation $U = a \cdot I_2 + b \cdot J$, where $a = (U)_{00} \in \mathbb{C}$

(the a -channel) and $b = (U)_{01} \in \mathbb{C}$ (the b -channel), with $|a|^2 + |b|^2 = 1$, the central new result of this paper — the projector identification of the b -channel — previews Proposition 3.1:

Proposition 3.1 (Off-Diagonal Rank-1 Collapse). For every $U_t, U_0 \in \text{SU}(2)$ with $E = I_2 \otimes U \otimes I_2$,

$$\text{Re Tr} \left[P_+^{\text{LW}} E_t P_-^{\text{LW}} E_0^\dagger \right] = 4 \text{Re}[b_t \bar{b}_0],$$

where $b_t = (U_t)_{01}$. *The new content is the projector identification: $P_\pm^{\text{LW}}$ selects the b -channel, giving Δ_{cross} an independent algebraic generator for the first time.*

From Proposition 3.1, together with Corollary 3.1 (Sec. 3.2) and Proposition 1 of Paper 4 [[P4]], the following results are established:

- **Theorem 3.1** (Sec. 3.3): $G_{\text{diag}}(t) + G_{\text{cross}}(t) = G_{\text{full}}(t)$ holds *exactly* for every $\text{SU}(2)$ gauge configuration and every timeslice t , where $G_{\text{cross}}(t) = 8 \sum \text{Re}[b_t \bar{b}_0]$. Theorem 3.1 simultaneously confirms that $G_{\text{cross}} \equiv \Delta_{\text{cross}}$.
- **Corollary 6.1** (Sec. 6.2, Two-Observable Decomposition): the pair $\{G_{\text{diag}}, G_{\text{cross}}\}$ constitutes an exact algebraic decomposition of G_{full} in the $\text{Cl}(0, 2) \cong \mathbb{H}$ framework.
- **Proposition 5.1** (Sec. 5.2): the rank-1 collapse is intrinsic to the fundamental $\text{SU}(2)$ representation, independent of the specific tensor-product embedding.
- **Proposition 7.1** (Sec. 7.2): the b -channel is a necessary algebraic conduit for any gauge-link correlator sensitive to $W^\pm$ at $O(g)$.
- **Proposition 8.1** (Sec. 8.1): Δ_{cross} is not gauge-invariant. In Landau gauge, $\Delta_{\text{cross}}(t)/8V \approx +(g^2/2) C_{W^\pm}(t)$ at $O(g^2)$.

The bridge between Paper 4 and Paper 5 is explicit: $G_{\text{cross}}(t) \equiv \Delta_{\text{cross}}(t)$, verified numerically to $\leq 1.14 \times 10^{-13}$ across all four ensembles (checks D7a–D7c, D8 $\times$ 4 β).

1.3 Scope Limits

This paper works entirely within the pure $\text{SU}(2)$ gauge sector. The following topics lie strictly outside its scope and are not addressed:

1. *No Higgs mechanism or electroweak symmetry breaking.* The gauge fields remain massless throughout; no scalar sector is introduced.
2. *No chiral fermions.* The Nielsen–Ninomiya obstruction [[NN1], [NN2]] is not overcome. Paper 5 contains no fermion action and makes no claim of implementing chiral symmetry on the lattice.
3. *No $\text{SU}(2)_L/\text{SU}(2)_R$ distinction.* The algebraic framework identifies G_{cross} as a b -channel probe, but does not select between left- and right-handed couplings. *The embedding does not, by itself, distinguish $\text{SU}(2)_L$ from $\text{SU}(2)_R$; this requires external fermionic input.*

4. *No dynamical symmetry-breaking signals.* Confinement and chiral condensate are not discussed; the ensembles are pure-gauge and thermalized in the absence of matter fields.

1.4 Paper Organisation

Section 2 establishes the algebraic framework: the $\text{Cl}(0, 2) \cong \mathbb{H}$ decomposition, the right $\mathbb{C}$ -module structure of the b -channel (Lemma 2.1), and the $\text{SU}(2)$ parametrisation. Section 3 proves Proposition 3.1, Corollary 3.1, and Theorem 3.1, establishing the two-observable decomposition algebraically and numerically. Section 4 interprets the a - and b -channels physically through the weak-coupling expansion. Section 5 proves embedding universality (Proposition 5.1) and the universal b -channel blindness of diagonal projectors (Lemma 5.1). Section 6 assembles the two-observable decomposition into Corollary 6.1. Section 7 examines the connection to $\text{SU}(2)_L$ through channel assignment (Proposition 7.1) and the conditional prediction for fermionic observables. Section 8 proves gauge non-invariance of Δ_{cross} (Proposition 8.1, Lemma 8.1) and develops the Landau gauge prescription. Section 9 synthesises the results in discussion, including five conclusions and seven open questions. Section 10 presents the conclusions.

2 Algebraic Framework: a/b -Channel Decomposition

2.1 The $\text{Cl}(0, 2) \cong \mathbb{H}$ Decomposition

The Clifford algebra $\text{Cl}(0, 2)$ is generated by two anticommuting orthonormal vectors $\hat{i}, \hat{j}$ satisfying $\hat{i}^2 = -1$, $\hat{j}^2 = -1$, and $\hat{i}\hat{j} = -\hat{j}\hat{i}$. The product $\hat{i}\hat{j}$ squares to -1 and anticommutes with both $\hat{i}$ and $\hat{j}$, giving the dimension count $1 + 2 + 1 = 4$ graded basis elements

$$\{1, \hat{i}, \hat{j}, \hat{i}\hat{j}\}.$$

As an associative algebra over $\mathbb{R}$, $\text{Cl}(0, 2)$ is isomorphic to the quaternion algebra $\mathbb{H}$ via the identification

$$1 \leftrightarrow 1, \quad \hat{i} \leftrightarrow i, \quad \hat{j} \leftrightarrow j, \quad \hat{i}\hat{j} \leftrightarrow k.$$

The complex subalgebra $\mathbb{C} \subset \text{Cl}(0, 2)$ is the algebra of elements of the form $\alpha + \beta\hat{i}$ with $\alpha, \beta \in \mathbb{R}$; the imaginary unit of this $\mathbb{C}$ is identified with $\hat{i}$.

The decomposition relevant for the present paper is the right- $\mathbb{C}$ module direct sum

$$\text{Cl}(0, 2) = \mathbb{C} \oplus \mathbb{C}\hat{j}, \tag{2.1}$$

where $\mathbb{C}\hat{j}$ is the right $\mathbb{C}$ -module of elements $b\hat{j}$ with $b \in \mathbb{C}$. The action of $\mathbb{C}$ on this module is from the right: $(b\hat{j}) \cdot c = b(\hat{j}c) = b\bar{c}\hat{j}$ for any $b, c \in \mathbb{C}$, using the anticommutation $\hat{j}c = \bar{c}\hat{j}$. The decomposition (2.1) is the algebraic foundation of the two-observable decomposition.

2.2 $\mathbb{C}\hat{j}$ is a Right $\mathbb{C}$ -Module

The structure $\mathbb{C}\hat{j}$ is precisely a right $\mathbb{C}$ -module, not a left $\mathbb{H}$ -module.

Lemma 2.1. *For any $b \in \mathbb{C}$ and $c \in \mathbb{C}$, defining the right action of $\mathbb{C}$ on the formal expression $b\hat{j}$ by*

$$(b\hat{j}) \cdot c := b(\hat{j}c) = b\bar{c}\hat{j} \quad (\text{using } \hat{j}c = \bar{c}\hat{j}),$$

the set $\mathbb{C}\hat{j} := \{b\hat{j} : b \in \mathbb{C}\}$ forms a one-dimensional right $\mathbb{C}$ -module under this action.

Properties RM1–RM5 are verified by direct expansion using the anticommutation rule $\hat{j}c = \bar{c}\hat{j}$ (checks D2a–d). The verification is summarised below:

Property	Statement	Verified
RM1	$((b_1 + b_2)\hat{j}) \cdot c = (b_1\hat{j} \cdot c) + (b_2\hat{j} \cdot c)$	✓
RM2	$(b\hat{j}) \cdot (c_1 + c_2) = (b\hat{j} \cdot c_1) + (b\hat{j} \cdot c_2)$	✓
RM3	$((b\hat{j}) \cdot c_1) \cdot c_2 = (b\hat{j}) \cdot (c_1c_2)$	✓
RM4	$(b\hat{j}) \cdot 1 = b\hat{j}$	✓
RM5	One-dimensional: spanned by $\{\hat{j}\}$ over $\mathbb{C}$	✓

The verification establishes that $\mathbb{C}\hat{j}$ is the formal one-dimensional right $\mathbb{C}$ -module, isomorphic to $\mathbb{C}$ as a $\mathbb{C}$ -vector space but with the scalar action carrying a complex conjugation.

This module structure is preserved exactly under the parametrisation $U = a \cdot I_2 + b \cdot J$ in Sec. 2.3. The b -channel of the $SU(2)$ decomposition inherits the right $\mathbb{C}$ -module structure: $b \in \mathbb{C}$ is the module coefficient, J is the analog of $\hat{j}$.

Per the standing correction: this is a right $\mathbb{C}$ -module, *not* a left $\mathbb{H}$ -module. Earlier references to a left $\mathbb{H}$ -module structure are incorrect; the correct algebraic identity is the right $\mathbb{C}$ -module decomposition (2.1).

2.3 $SU(2)$ Parametrisation

Every $SU(2)$ matrix admits the parametrisation

$$U = \begin{pmatrix} a & b \\ -b^* & a^* \end{pmatrix} = a \cdot I_2 + b \cdot J \quad (|a|^2 + |b|^2 = 1), \quad (2.2)$$

where:

- $a = (U)_{00} \in \mathbb{C}$, called the *a-channel*;
- $b = (U)_{01} \in \mathbb{C}$, called the *b-channel*.

The matrix J in (2.2) is the 2×2 representation of the quaternion unit $\hat{j}$:

$$J = i\sigma_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad (2.3)$$

satisfying $J^2 = -I_2$ and $J^\dagger = -J$, consistent with the Clifford relation $\hat{j}^2 = -1$.

When the $\text{Cl}(0, 2) \cong \mathbb{H}$ decomposition (2.1) is enforced, b is more naturally written as $b\hat{j} \in \mathbb{C}\hat{j}$, and the correlator bilinears acquire the right $\mathbb{C}$ -module interpretation:

$$\text{Re}[b_t \bar{b}_0] \longleftrightarrow \text{“right } \mathbb{C}\text{-module bilinear in } \mathbb{C}\hat{j}\text{”}.$$

This identification is what gives G_{cross} its algebraic identity as the b -channel observable.

The fundamental $\text{SU}(2)$ representation in this paper is 8-dimensional, embedded as $U \in M_2(\mathbb{C}) \rightarrow I_2 \otimes U \otimes I_2 \in M_8(\mathbb{C})$. All assertions in this paper are independent of the choice of tensor-product ordering; see Proposition 5.1 in Sec. 5.2 for the precise universality statement.

For verification, all bench tests use:

- $N = 6^3 \times 6$ lattice ($V = 6^4 = 1296$);
- $\beta_{\text{eff}} \in \{5.0, 6.0, 9.0, 12.0\}$;
- 20 thermalized configurations per ensemble;
- Standard heat-bath update; eigenspace projectors $P_{\pm}^{\text{LW}}$ from Paper 4 [[P4]].

A complete map of the algebraic identifications is given in Table 1.

Table 1: Algebraic identifications and physical role assignments for the $\text{Cl}(0, 2) \cong \mathbb{H} \leftrightarrow \text{SU}(2) \leftrightarrow M_2(\mathbb{C})$ chain. The fundamental $\text{SU}(2)$ representation (8-dim) provides the chain; each level inherits the right- $\mathbb{C}$ -module structure of the b -channel.

Object	Algebraic ID	Matrix Form	Role
$\hat{i}$	imaginary unit	$i\sigma_3$	Cartan e
$\hat{j}$	second generator	$i\sigma_2$	off-diagon
$\hat{i}\hat{j}$	pseudoscalar	$-i\sigma_1$	volume f
1	identity	I_2	scalar
$a \in \mathbb{C}$	a -channel	$(U)_{00}$	diagonal
$b \in \mathbb{C}$	b -channel	$(U)_{01}$	off-diagon
$P_{\pm}^{\text{LW}}$	rank-1 T_3 -eigenprojector	$(I_2 \pm \sigma_3)/2$	eigenspac
Π_+	rank-1 fundamental	$(I_2 + \sigma_3)/2$	a -channe
Π_-	rank-1 fundamental	$(I_2 - \sigma_3)/2$	b -channe
$G_{\text{diag}}(t)$	diagonal observable	$\text{Re}(\text{Tr}[P_+^{\text{LW}} E_t P_+^{\text{LW}} E_0^\dagger] + \text{Tr}[P_-^{\text{LW}} E_t P_-^{\text{LW}} E_0^\dagger])$	a -channe
$G_{\text{cross}}(t)$	off-diagonal observable	$\text{Re}(\text{Tr}[P_+^{\text{LW}} E_t P_-^{\text{LW}} E_0^\dagger] + \text{Tr}[P_-^{\text{LW}} E_t P_+^{\text{LW}} E_0^\dagger]) = 8 \sum \text{Re}[b_t \bar{b}_0]$ (Eq. 3.5)	b -channe

3 Off-Diagonal Rank-1 Collapse and Two-Observable Completeness

This section proves Proposition 3.1 and Corollary 3.1, then assembles the two-observable decomposition Theorem 3.1.

3.1 Off-Diagonal Rank-1 Collapse

Proposition 3.1 (Off-Diagonal Rank-1 Collapse). *For every $U_t, U_0 \in \text{SU}(2)$ with $E = I_2 \otimes U \otimes I_2$:*

$$\text{Re Tr} \left[P_+^{\text{LW}} E_t P_-^{\text{LW}} E_0^\dagger \right] = 4 \text{Re}[b_t \bar{b}_0], \quad (3.1)$$

where $b_t = (U_t)_{01}$.

Proof. Recall the rank-1 collapse identity: for any $A \in M_2(\mathbb{C})$,

$$\Pi_r A \Pi_{r'} = (A)_{rr'} \Pi_r \Pi_{r'} = (A)_{rr'} \delta_{rr'} \Pi_r \quad (\text{no sum}).$$

Applied to $A = E_t = I_2 \otimes U_t \otimes I_2$ with Π_+, Π_- the rank-1 fundamental projectors acting on the central $\text{SU}(2)$ factor:

$$P_+^{\text{LW}} E_t P_-^{\text{LW}} = (I_2 \otimes U_t \otimes I_2)_{01} (I_2 \otimes \Pi_+ \Pi_- \otimes I_2) = (U_t)_{01} \cdot (\text{rank-1 vanishing term}) = b_t (\Pi_+ \Pi_- \text{ in the t})$$

Tracing against $E_0^\dagger$:

$$\text{Tr} \left[P_+^{\text{LW}} E_t P_-^{\text{LW}} E_0^\dagger \right] = b_t \cdot (E_0^\dagger)_{10} = b_t \bar{b}_0,$$

using $(E_0^\dagger)_{10} = ((U_0)_{01})^* = \bar{b}_0$. Taking the real part:

$$\text{Re Tr} \left[P_+^{\text{LW}} E_t P_-^{\text{LW}} E_0^\dagger \right] = \text{Re}[b_t \bar{b}_0].$$

The factor of 4 arises from the $I_2 \otimes (\cdot) \otimes I_2$ tensor product:

$$\text{Re Tr}_{8 \times 8} \left[P_+^{\text{LW}} E_t P_-^{\text{LW}} E_0^\dagger \right] = 4 \text{Re Tr}_{2 \times 2} \left[\Pi_+ U_t \Pi_- U_0^\dagger \right] = 4 \text{Re}[b_t \bar{b}_0]. \quad \square$$

The verification was performed across all four ensembles using checks D1, D2a–d, D3a–d, D4a–h, D5, D6, D7a–c with results listed in Table 2.

3.2 Orientation Symmetry of the Off-Diagonal Collapse

Corollary 3.1 (Orientation Symmetry). *For every $U_t, U_0 \in \text{SU}(2)$:*

$$\text{Re Tr} \left[P_-^{\text{LW}} E_t P_+^{\text{LW}} E_0^\dagger \right] = \text{Re Tr} \left[P_+^{\text{LW}} E_t P_-^{\text{LW}} E_0^\dagger \right] = 4 \text{Re}[b_t \bar{b}_0]. \quad (3.2)$$

Proof. Both reduce to the same expression upon application of Proposition 3.1 to the conjugate pair $\{P_+^{\text{LW}}, P_-^{\text{LW}}\}$. The orientation reversal exchanges $\text{Tr}[X^\dagger] = \text{Tr}[X]^*$ for the real part:

$$\text{Re Tr} \left[P_-^{\text{LW}} E_t P_+^{\text{LW}} E_0^\dagger \right] = \text{Re Tr} \left[\left(P_-^{\text{LW}} E_t P_+^{\text{LW}} E_0^\dagger \right)^\dagger \right] = \text{Re Tr} \left[E_0 P_+^{\text{LW}} E_t P_-^{\text{LW}} \right].$$

Cycling and using $P_r^{\text{LW}2} = P_r^{\text{LW}}$:

$$= \text{Re Tr} \left[P_+^{\text{LW}} E_t P_-^{\text{LW}} E_0 \right] = (E_t^\dagger)_{01} (E_0)_{10}^* = (\bar{b}_t)(\bar{b}_0)^* = \bar{b}_t b_0 = (b_t \bar{b}_0)^*.$$

So $\text{Re Tr} \left[P_-^{\text{LW}} E_t P_+^{\text{LW}} E_0^\dagger \right] = \text{Re}[\bar{b}_t b_0] = \text{Re}[b_t \bar{b}_0]$. $\square$

The orientation symmetry confirms that the off-diagonal bilinear is a single channel (not two), justifying the single G_{cross} observable.

Table 2: Verification checks for Proposition 3.1. All 18 check IDs (51 instances across D1–D18) pass at tolerance $\leq 10^{-13}$ or better.

Check	Result	Tolerance
D1	Rank-1 collapse identity $\Pi_r A \Pi_r = (A)_{rr} \Pi_r$ — algebraic	exact
D2a–d	$\hat{j} c = \bar{c} \hat{j}$ for $c = 1, \hat{i}, \beta, \hat{i} \cdot \beta$	$\leq 10^{-15}$
D3a–d	$(b \hat{j}) \cdot c = b \cdot \bar{c} \cdot \hat{j}$ verification	$\leq 10^{-15}$
D4a–h	Direct rank-1 identity in 8-dim: $\Pi_r E \Pi_{r'} = \delta_{rr'} (M)_{rr}$	$\leq 10^{-13}$
D5	Off-diagonal collapse $\Pi_+ E \Pi_- = (E)_{01} \Pi_+ \Pi_-$	$\leq 10^{-13}$
D6	Off-diagonal rank-1 collapse identity (8×8) : $\text{Tr}[\Pi_+ E \Pi_- E'^\dagger] = 4 (U)_{01} (U')_{01}^*$	$\leq 10^{-13}$
D7a–c	$\text{Re Tr}[P_+^{\text{LW}} E_t P_-^{\text{LW}} E_0^\dagger] - 4 \text{Re}[b_t \bar{b}_0] = 0$ across ensembles	$\leq 10^{-13}$

3.3 Two-Observable Completeness

Theorem 3.1 (Two-Observable Completeness). *For every SU(2) gauge field configuration and every t :*

$$G_{\text{diag}}(t) + G_{\text{cross}}(t) = G_{\text{full}}(t), \quad (3.3)$$

where

$$G_{\text{diag}}(t) = 8 \sum \text{Re}[a_t \bar{a}_0], \quad (3.4)$$

$$G_{\text{cross}}(t) = 8 \sum \text{Re}[b_t \bar{b}_0], \quad (3.5)$$

$$G_{\text{full}}(t) = \sum \text{Re Tr}[E_t E_0^\dagger]. \quad (3.6)$$

The sums are over space and over both the (configuration, t) pairs.

Proof. Expand $E_t E_0^\dagger$ using the SU(2) parametrisation (2.2):

$$\text{Tr}[E_t E_0^\dagger] = \text{Tr}[(a_t I_2 + b_t J)(\bar{a}_0 I_2 - \bar{b}_0 J)].$$

Inserting completeness $\Pi_+ + \Pi_- = I_8$ (or I_2 in the central factor):

$$E_t E_0^\dagger = (\Pi_+ + \Pi_-) E_t E_0^\dagger (\Pi_+ + \Pi_-) = \Pi_+ E_t E_0^\dagger \Pi_+ + \Pi_+ E_t E_0^\dagger \Pi_- + \Pi_- E_t E_0^\dagger \Pi_+ + \Pi_- E_t E_0^\dagger \Pi_-.$$

Tracing and taking real parts: the two diagonal blocks $(\Pi_+ \cdots \Pi_+, \Pi_- \cdots \Pi_-)$ each contribute $4 \text{Re}[a_t \bar{a}_0]$ by the diagonal collapse identity (Proposition 1 of Paper 4 [[P4]]), summing to $8 \text{Re}[a_t \bar{a}_0]$. The two off-diagonal blocks $(\Pi_+ \cdots \Pi_-, \Pi_- \cdots \Pi_+)$ each contribute $4 \text{Re}[b_t \bar{b}_0]$ by Proposition 3.1 and Corollary 3.1, summing to $8 \text{Re}[b_t \bar{b}_0]$. Therefore:

$$\text{Re Tr}[E_t E_0^\dagger] = 8 \text{Re}[a_t \bar{a}_0] + 8 \text{Re}[b_t \bar{b}_0].$$

Summing over space gives $G_{\text{full}} = G_{\text{diag}} + G_{\text{cross}}$. □

The simultaneous decomposition was verified numerically: check D7a–c gives $|G_{\text{diag}} + G_{\text{cross}} - G_{\text{full}}| \leq 1.78 \times 10^{-15}$ across all four ensembles, confirming Theorem 3.1 to the precision of the simulation; check $D8 \times 4\beta$ additionally gives $|G_{\text{cross}} - \Delta_{\text{cross}}| \leq 1.14 \times 10^{-13}$, confirming the bridge identification $G_{\text{cross}} \equiv \Delta_{\text{cross}}$.

Theorem 3.1 simultaneously establishes $G_{\text{cross}} \equiv \Delta_{\text{cross}}$: the algebraically defined G_{cross} equals the subtractionally defined Δ_{cross} . This resolves the structural gap of Paper 4 [[P4]], §4.3: Δ_{cross} is not merely a residual, but the b -channel bilinear in the right $\mathbb{C}$ -module decomposition.

4 Physical Interpretation: W^3 vs. $W^\pm$

4.1 Weak-Coupling Expansion of the Link Variable

Setting $U_\mu(x) = \exp(ig a_{\text{lat}} A_\mu^c T^c)$ and expanding to leading order in the coupling g , with generators $T^c = \sigma^c/2$ in the fundamental representation:

$$a = (U)_{00} = 1 + \frac{ig}{2} A_\mu^3 + O(g^2), \quad (4.1)$$

$$b = (U)_{01} = \frac{ig}{2} (A_\mu^1 - i A_\mu^2) + O(g^2), \quad (4.2)$$

$$b = (U)_{01} = \frac{ig}{\sqrt{2}} W_\mu^+ + O(g^2), \quad W_\mu^+ = \frac{A_\mu^1 - i A_\mu^2}{\sqrt{2}}. \quad (4.3)$$

4.2 Channel-to-Boson Identification

Equations (4.1)–(4.3) establish a direct correspondence at $O(g)$:

$$G_{\text{diag}} \longleftrightarrow A_\mu^3 \quad (\text{neutral gauge boson sector}), \quad G_{\text{cross}} \longleftrightarrow W_\mu^\pm \quad (\text{charged gauge boson sector}).$$

This identification holds to $O(g)$ in the matrix elements and to $O(g^2)$ in the bilinears. The channel separation— a diagonal, b off-diagonal—is an exact property of the $\text{SU}(2)$ matrix structure and holds for all g .

Table 3: Weak-coupling field content of the $\text{SU}(2)$ link variable. The third column uses A^3 for the leading-order neutral field.

Matrix entry	SU(2) param	Leading order in g	Gauge boson
$(U)_{00}$	a	$1 + (ig/2) A^3$	Neutral (W^3)
$(U)_{01}$	b	$(ig/\sqrt{2}) W^+$	Charged (W^+)
$(U)_{10}$	$-b^*$	$(ig/\sqrt{2}) W^-$	Charged (W^-)
$(U)_{11}$	a^*	$1 - (ig/2) A^3$	Neutral (W^3)

4.3 $O(g^2)$ Calculation of the Channel Bilinears

$$\operatorname{Re}[a_t \bar{a}_0] = 1 + \frac{g^2}{4} A_t^3 A_0^3 + O(g^3), \quad (4.4)$$

$$\operatorname{Re}[b_t \bar{b}_0] = \frac{g^2}{4} (A_t^1 A_0^1 + A_t^2 A_0^2) + O(g^3). \quad (4.5)$$

Equation (4.5) shows that $\operatorname{Re}[b_t \bar{b}_0]$ is $O(g^2)$: it vanishes as $g \rightarrow 0$ (free field). The measured ratio $\Delta_{\text{cross}}/G_{\text{full}} \in [0.72, 2.33]$ at $\beta_{\text{eff}} \in \{5.0, 6.0, 9.0, 12.0\}$ [[P4]] reflects non-perturbative finite-coupling contributions where the $O(g^2)$ approximation is not quantitatively valid.

Remark 4.1 (D10b — Slope of b -Channel Weak-Coupling Error). The log-log slope of the b -channel approximation error vs. ε is 3.000, not 2 (check D10b, flagged and closed). This arises because $H^2 \propto I_2$, so $(H^2)_{01} = 0$, and the Taylor expansion of $\sin(\varepsilon \|H\|)$ has no ε^2 term in the off-diagonal entry; the leading error is $O(\varepsilon^3)$. This is the sin-Taylor argument. It is distinct from the BCH/ $[T_+, T_-] \propto T_3$ argument (check D17), which explains channel purity, not slope. Slope = 3 confirms approximation quality; channel identity is established algebraically by Proposition 3.1, Lemma 5.1, and checks D3c–D3d. Checks D11a, D11b, D12b, D13a verify the a -channel slope, monotone $|R(\beta)|$ at $t = 5$, and cold-start $\langle |b|^2 \rangle$.

4.4 What the Channel Identification Does Not Imply

- (OS4.1) *No W/Z mass ratio.* No Weinberg angle θ_W , no $U(1)_Y$, no Higgs sector.
- (OS4.2) *No L/R distinction.* The $SU(2)$ link variable U transforms identically regardless of the handedness of any matter field; chirality is encoded in the matter sector only.
- (OS4.3) *No Higgs mechanism.* The gauge fields in the present $SU(2)$ pure-gauge theory are massless.
- (OS4.4) *No electroweak mixing.* The hypercharge gauge group $U(1)_Y$ is absent; A^3 should not be identified with the physical Z boson or photon prior to mixing.

5 Embedding Universality

5.1 The Core Rank-1 Collapse Lemma

Core Lemma (Rank-1 Collapse). *For any $U \in \operatorname{End}(\mathbb{C}^2)$ and any rank-1 projector $\Pi_r = |r\rangle\langle r|$:*

$$\Pi_r U \Pi_r = (U)_{rr} \cdot \Pi_r. \quad (5.1)$$

Proof. $\Pi_r U \Pi_r = |r\rangle\langle r| U |r\rangle\langle r| = (U)_{rr} |r\rangle\langle r| = (U)_{rr} \Pi_r.$ $\square$

5.2 Rank-1 Collapse Universality

We consider three natural embeddings of $U \in \text{SU}(2)$ into $\text{End}(\mathbb{C}^8)$:

$$\begin{aligned} \text{Case A:} \quad & E = U \otimes I_4, \\ \text{Case B:} \quad & E = I_4 \otimes U, \\ \text{Case C:} \quad & E = I_2 \otimes U \otimes I_2 \quad (\text{used in Papers 4 and 5}). \end{aligned}$$

Proposition 5.1 (Rank-1 Collapse Universality). *For each embedding Case A, B, C and for $r \in \{+, -\}$:*

$$\text{Re Tr}[P_r E_t P_r E_0^\dagger] = 4 \text{Re}[(U_t)_{rr} \cdot \overline{(U_0)_{rr}}]. \quad (5.2)$$

Proof. In each case, P_r factors as a tensor product with Π_r acting on the U -component and identity operators on the spectator components. Tensor multiplicativity gives

$$\text{Tr}[P_r E_t P_r E_0^\dagger] = \text{Tr}(I_{\text{spec}}) \cdot \text{Tr}[\Pi_r U_t \Pi_r U_0^\dagger],$$

where $\text{Tr}(I_{\text{spec}}) = 4$ in all three cases. Applying the Core Lemma (5.1) and evaluating the trace gives $4 \text{Re}[(U_t)_{rr} \overline{(U_0)_{rr}}]$. $\square$

Verified for Cases A and B by checks D14⁺, D14[−], D15⁺, D15[−] on 50 random SU(2) pairs each. Residuals are 2.22×10^{-16} (1 ULP) in all cases. These are type (b) independent computations: the LHS uses an 8×8 Kronecker-product trace, the RHS accesses the 2×2 matrix element directly; the two code paths share no intermediate computation.

5.3 Intrinsic Nature of the Decomposition

Corollary 5.1 (Universality of the Two-Observable Structure). *The rank-1 collapse formula $\Pi_r U \Pi_r = (U)_{rr} \Pi_r$ —and hence the a/b-channel separation and the two-observable decomposition of Theorem 3.1—is intrinsic to the fundamental spin-1/2 representation of SU(2). It is not an artifact of the tensor-product embedding $I_2 \otimes U \otimes I_2$ adopted in Paper 4. Escape condition: reducible representations with degenerate T_3 eigenspaces avoid rank-1 collapse; no such representations appear in this series.*

5.4 b-Channel Blindness is Universal

Lemma 5.1 (b-Channel Blindness Universality). *For every embedding Case A, B, C, the diagonal projector correlator*

$$\text{Re Tr}[P_r E_t P_r E_0^\dagger] = 4 \text{Re}[(U_t)_{rr} \cdot \overline{(U_0)_{rr}}]$$

is independent of $b = (U)_{01}$ for every $U_t, U_0 \in \text{SU}(2)$.

Proof. The RHS of (5.2) contains only $(U)_{rr} = a$ (for $r = +$) or a^* (for $r = -$), and no occurrence of $b = (U)_{01}$. Independence follows from Proposition 5.1. $\square$

Check D16 provides a consistency check over 800 evaluations, finding $\text{std} = 0.00 \times 10^{+00}$ for the dependence on b . This result is guaranteed by construction (b is algebraically absent from the formula) and should be read as a consistency check rather than an independent verification. The substantive verification is established by Proposition 5.1 and the Core Lemma (5.1), and confirmed by the independent code-path comparison of D14/D15.

Table 4: Three-embedding comparison. Each embedding yields the same diagonal collapse formula and the same b -blindness, differing only in how the spectator dimensions assemble to $\text{Tr}(I_{\text{spec}}) = 4$.

Case	Embedding	Formula	b -blind?	Spectator dim
A	$U \otimes I_4$	$4 \text{Re}[(U_t)_{rr} \overline{(U_0)_{rr}}]$	Yes	$\text{Tr}(I_4) = 4$
B	$I_4 \otimes U$	$4 \text{Re}[(U_t)_{rr} \overline{(U_0)_{rr}}]$	Yes	$\text{Tr}(I_4) = 4$
C	$I_2 \otimes U \otimes I_2$	$4 \text{Re}[(U_t)_{rr} \overline{(U_0)_{rr}}]$	Yes	$\text{Tr}(I_2)^2 = 4$

6 The Two-Observable Decomposition Structure

6.1 Setup and Definitions

Theorem 3.1 (Sec. 3.3) establishes that the full gauge-link correlator decomposes exactly into two independent bilinear contributions. The defining equations are recalled below; per Q1 the original (6.1a)–(6.1c) sub-letter triplet has been split into three independently numbered equations:

$$G_{\text{diag}}(t) = 8 \sum_{\text{links}} \text{Re}[a_t \bar{a}_0], \quad (6.1)$$

$$G_{\text{cross}}(t) = 8 \sum_{\text{links}} \text{Re}[b_t \bar{b}_0], \quad (6.2)$$

$$G_{\text{full}}(t) = G_{\text{diag}}(t) + G_{\text{cross}}(t). \quad (6.3)$$

Equation (6.3) is exact for every SU(2) gauge configuration and every timeslice t (Theorem 3.1).

6.2 Two-Observable Decomposition

Corollary 6.1 (Two-Observable Decomposition in the Cl(0, 2) $\cong \mathbb{H}$ Framework). *The following holds exactly for every SU(2) field configuration and every t :*

- (i) **Exactness** [NEW — PROVED AS THEOREM 3.1]: $G_{\text{diag}}(t) + G_{\text{cross}}(t) = G_{\text{full}}(t)$.
- (ii) **a -Channel Faithfulness** [RECALLED — PROPOSITION 1 OF PAPER 4 [[P4]], §4.2]: G_{diag} depends only on a_t and $\bar{a}_0$.

(iii) ***b*-Channel Faithfulness** [NEW — PROVED AS PROPOSITION 3.1]: G_{cross} depends only on b_t and $\bar{b}_0$.

Corollary 6.1 follows immediately from Theorem 3.1, Proposition 1 of Paper 4 [[P4]] (§4.2), and Proposition 3.1. Parts (i) and (iii) are new results established in this paper; part (ii) assembles Proposition 1 of Paper 4 for completeness of statement.

Remark 6.1 (Parametric completeness). The pair $\{a, b\}$ with $|a|^2 + |b|^2 = 1$ parametrises $\text{SU}(2) \cong S^3$. This is a conjunction of (ii), (iii), and the unitarity constraint $|a|^2 + |b|^2 = 1$ (labelled (E4) in Paper 4 [[P4]], retained here for cross-paper consistency) —not an independent result—but it justifies describing the decomposition as complete.

Remark 6.2 (Weak-coupling channel identification). At leading order in g : $a = 1 + (ig/2)A^3 + O(g^2)$ and $b = (ig/\sqrt{2})W^+ + O(g^2)$ (Eqs. (4.1), (4.3)). Therefore G_{diag} probes A^3 and G_{cross} probes $W^\pm$. This identification is perturbative—valid to $O(g^2)$ in the bilinears—and does not constitute part of the exact decomposition of Corollary 6.1.

Proof of Corollary 6.1. Immediate from Theorem 3.1 (Sec. 3.3), Proposition 1 of Paper 4 [[P4]] (§4.2), and Proposition 3.1 (Sec. 3.1). $\square$

6.3 Why $\{G_{\text{diag}}, G_{\text{cross}}\}$ and Not $\{G_+, G_-\}$

By Corollary 1 of Paper 4 [[P4]], §4.4, $G_+(t) = G_-(t)$ identically for every $U \in \text{SU}(2)$. Their sum is $G_+ + G_- = 2G_+ = G_{\text{diag}}$, and the pair $\{G_+, G_-\}$ carries exactly one independent observable—the a -channel bilinear—and no information about b . The apparent two-dimensional structure collapses to a single observable by Corollary 1 of Paper 4, leaving the entire b -channel inaccessible.

The pair $\{G_{\text{diag}}, G_{\text{cross}}\}$, by contrast, provides two genuinely independent observables: G_{diag} is faithful to a (part (ii)) and G_{cross} is faithful to b (part (iii)), with the two channels linked only by the unitarity constraint $|a|^2 + |b|^2 = 1$, which imposes no algebraic relation between $\text{Re}[a_t \bar{a}_0]$ and $\text{Re}[b_t \bar{b}_0]$ across timeslices.

6.4 Sign Non-Definiteness

Proposition 6.1 (Sign Non-Definiteness). $G_{\text{diag}}(t)$ and $G_{\text{cross}}(t)$ are individually not sign-definite per configuration:

- (a) $G_{\text{diag}} < 0$ is possible. Example: $a_t = -1/\sqrt{2} + i\varepsilon$, $a_0 = 1/\sqrt{2}$; then $\text{Re}[a_t \bar{a}_0] = -1/2 < 0$.
- (b) $G_{\text{cross}} < 0$ is possible. Example: $b_t = -1/\sqrt{2}$, $b_0 = 1/\sqrt{2}$; then $\text{Re}[b_t \bar{b}_0] = -1/2 < 0$.
- (c) G_{full} is not constrained positive per link by unitarity alone.

Proof. Parts (a) and (b) follow by direct substitution; both examples satisfy $|a|^2 + |b|^2 = 1$. Part (c) follows from (a), (b), and Eq. (6.3). The numerical observation $\Delta_{\text{cross}}/G_{\text{full}} > 1$ for some (β, t) pairs [[P4]] implies $G_{\text{diag}} < 0$ in those cases. $G_{\text{cross}} < 0$ was observed in 37–62% of (configuration, t) pairs [[P4]]. $\square$

6.5 Unitarity Constraint and Functional Independence

Corollary 6.2 (Unitarity Constraint). *The $SU(2)$ constraint $|a|^2 + |b|^2 = 1$ holds per link per timeslice. It imposes no algebraic relation between $\text{Re}[a_t \bar{a}_0]$ and $\text{Re}[b_t \bar{b}_0]$ across distinct timeslices $t \neq 0$. The two channels are functionally independent observables.*

Proof. The constraint $|a_t|^2 + |b_t|^2 = 1$ bounds moduli individually but places no constraint on the phases $\arg(a_t \bar{a}_0)$ or $\arg(b_t \bar{b}_0)$, which determine the real parts of the bilinears. For fixed $t = 0$ reference link and varying $t > 0$, the values of $\text{Re}[a_t \bar{a}_0]$ and $\text{Re}[b_t \bar{b}_0]$ range independently over $[-1, 1]$. $\square$

7 Connection to $SU(2)_L / SU(2)_R$

7.1 Channel Assignment at $O(g)$

The three $SU(2)$ generators in the fundamental representation are $T^3 = \text{diag}(1, -1)/2$, $T_+ = \sigma_+/2$, $T_- = \sigma_-/2$. The key point is that T^3 is purely diagonal and $T_\pm$ are purely off-diagonal. Consequently, at $O(g)$ in $U = I + ig a_{\text{lat}} A_\mu^c T^c + O(g^2)$:

- $g A_\mu^3 T^3$ contributes only to $(U)_{00}$ and $(U)_{11}$ (a -channel only).
- $g W_\mu^+ T_-$ and $g W_\mu^- T_+$ contribute only to $(U)_{01}$ and $(U)_{10}$ (b -channel only).

This separation is exact at $O(g)$. Checks D17-R1, D17-R2, D17-R3 verify: R1 confirms $\text{Im}(a_\varepsilon - 1)/\varepsilon \rightarrow \alpha^3/2$ (only $T^3 A^3$); R2 confirms $\partial(\text{Im } a)/\partial\alpha^1 = 0$ (no A^1 contamination); R3 confirms $\partial(\text{Im } a)/\partial\alpha^2 = 0$ (no A^2 contamination).

Table 5: Leading-order $SU(2)$ channel assignment. Each generator–field combination contributes to exactly one matrix element of U at $O(g)$, mapping cleanly onto either the a - or b -channel.

Coupling	Matrix element	Channel	$O(g)$ contribution
$T^3 A_\mu^3$	$(U)_{00} = a$	a -channel	$+(ig/2) A^3$
$T_- W_\mu^+$	$(U)_{01} = b$	b -channel	$+(ig/\sqrt{2}) W^+$
$T_+ W_\mu^-$	$(U)_{10} = -b^*$	b -channel	$+(ig/\sqrt{2}) W^-$
$T^3 A_\mu^3$	$(U)_{11} = a^*$	a -channel	$-(ig/2) A^3$

7.2 The b -Channel as Necessary Conduit

Proposition 7.1 (b -Channel Necessary Conduit). *(b1) **Necessity.** Any gauge-link correlator sensitive to $W^\pm$ at $O(g)$ must involve the b -channel. No diagonal projector correlator of the form $\text{Re } \text{Tr}[P_r E_t P_r E_0^\dagger]$ can access b (Lemma 5.1).*

*(b2) **Non-sufficiency.** The b -channel does not distinguish $SU(2)_L$ from $SU(2)_R$. The link variable U transforms identically under $SU(2)$ regardless of the handedness of any matter field. Chirality is encoded in the matter sector only.*

Proof. Proof of (b1). By Lemma 5.1, the diagonal projector correlator is independent of b for every embedding case. Since $W^\pm$ enter only through b at $O(g)$ (Table 5), no diagonal projector correlator can be sensitive to $W^\pm$. $G_{\text{cross}} = 8 \sum \text{Re}[b_t \bar{b}_0]$ is sensitive to $W^\pm$ by (4.3) and is the unique algebraic structure accessing b (Proposition 3.1).

Proof of (b2). The pure-gauge link variable carries no record of fermion handedness; this is determined by the transformation properties of fermion fields under Lorentz and gauge symmetry. $\square$

The embedding does not, by itself, distinguish $\text{SU}(2)_L$ from $\text{SU}(2)_R$; this requires external fermionic input.

7.3 Conditional Prediction for Fermionic Observables — Programme Target

This subsection describes a programme target for future work. It is not a result of Paper 5. Paper 5 contains no fermion action and makes no claim of testing the following prediction.

Suppose a fermionic propagator $S(x, y)$ is introduced on the same lattice, transforming in the fundamental representation of $\text{SU}(2)$. Define the cross-projected propagator:

$$S_{\text{cross}}(x, y) = P_+^{\text{LW}} S(x, y) P_-^{\text{LW}}. \quad (7.1)$$

The conditional prediction of the b -channel framework is:

- For an $\text{SU}(2)_L$ doublet fermion: $S_{\text{cross}} \neq 0$ (the doublet mixes isospin components via the $T_- W_\mu^+$ coupling).
- For an $\text{SU}(2)_R$ singlet fermion: $S_{\text{cross}} = 0$ (a singlet carries no $\text{SU}(2)$ index).

The Nielsen–Ninomiya obstruction [[NN1], [NN2]] states that no local, Hermitian, translation-invariant lattice fermion action can simultaneously be free of fermion doublers and chirally symmetric. Implementation requires domain wall [[Ka92]] or overlap fermions [[Ne98]] (Ginsparg–Wilson). This is designated Future Requirements F1–F4 and constitutes the entry point for Paper 6.

8 Gauge Properties of Δ_{cross} and the Landau Gauge Prescription

8.1 Proof of Gauge Non-Invariance

Lemma 8.1 (Transformation of b under Gauge Rotations). *Under $U' = g U h^\dagger$, with $g = \begin{pmatrix} \alpha & \beta \\ -\beta^* & \alpha^* \end{pmatrix}$ and $h = \begin{pmatrix} \gamma & \delta \\ -\delta^* & \gamma^* \end{pmatrix}$ elements of $\text{SU}(2)$, the transformed off-diagonal element is*

$$b' = (U')_{01} = \alpha \gamma b + \beta \gamma a^* - \alpha \delta a + \beta \delta \bar{b}. \quad (8.1)$$

Proof. Direct matrix multiplication of $(g U h^\dagger)_{01} = \sum_{j,k} (g)_{0j} (U)_{jk} (h^\dagger)_{k1}$, substituting the SU(2) parametrisations. *Counterexample:* $g_t = i\sigma_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, $h_t = I_2$ gives $b'_t = a_t^*$, demonstrating that b is not gauge-invariant. Verified on 50 random SU(2) pairs by check D18b-f (residual $0.00 \times 10^{+00}$). $\square$

Proposition 8.1 (Δ_{cross} is Not Gauge-Invariant). *Under the lattice gauge transformation $U_\mu(x) \rightarrow g(x) U_\mu(x) g^\dagger(x + \hat{\mu})$, with $g(x) \in \text{SU}(2)$, $\Delta_{\text{cross}}(t) = 8 \sum \text{Re}[b_t \bar{b}_0]$ changes.*

Proof. Under the gauge transformation with $g_t \neq g_0$, the transformed elements b'_t and b'_0 each depend on (g_t, g_0) through independent linear combinations of $a_t, b_t, a_t^*, \bar{b}_t$ and $a_0, b_0, a_0^*, \bar{b}_0$ respectively (Lemma 8.1). $\text{Re}[b'_t \bar{b}'_0] \neq \text{Re}[b_t \bar{b}_0]$ generically. *Counterexample:* $g_t = i\sigma_2, g_0 = I_2$ gives $b'_t = a_t^*, b'_0 = b_0$, so $\text{Re}[b'_t \bar{b}'_0] = \text{Re}[a_t^* \bar{b}_0] \neq \text{Re}[b_t \bar{b}_0]$ generically. $\square$

Check D18a (inverted) verifies gauge non-invariance on 50 configurations using Haar-random gauge transformations, finding min residual $1.07 \times 10^{-2} \gg 10^{-6}$. Check D18b-n verifies the specific counterexample. Scope: D18 is designed for Haar-random transforms; near-identity transforms ($g \approx I + i\varepsilon H$, $\varepsilon \lesssim 10^{-7}$) not tested. Proposition 8.1 proves non-invariance analytically for all transforms without restriction on the magnitude of g .

8.2 Landau Gauge Prescription

Since Δ_{cross} is gauge-dependent (Proposition 8.1), a gauge-fixing prescription is required. The standard choice for gluon propagator studies is Landau gauge, defined by the maximisation condition:

$$F_L[\{g(x)\}] = \text{Re} \sum_{x,\mu} \text{Tr}[U_\mu^g(x)] \longrightarrow \text{maximum over } \{g(x)\}, \quad (8.2)$$

where $U_\mu^g(x) = g(x) U_\mu(x) g^\dagger(x + \hat{\mu})$. The Landau condition $\partial_\mu A_\mu^a = 0$ is recovered in the continuum limit. The target comparison data are the SU(2) Landau-gauge gluon propagator computations of Cucchieri and Mendes [[CM]] and Bogolubsky et al. [[Bo]].

8.3 Formal Comparison to the $W^\pm$ Propagator

In Landau gauge, the $W^\pm$ propagator at zero spatial momentum is:

$$C_{W^\pm}(t) = \frac{1}{V} \sum_x \langle W_i^+(x, t) W_i^-(x, 0) \rangle.$$

Using the weak-coupling identification (4.3), $b \approx (ig/\sqrt{2}) W^+$, so $\bar{b}_0 \approx (-ig/\sqrt{2}) W_0^-$, giving

$$b_t \bar{b}_0 \approx (ig/\sqrt{2}) (-ig/\sqrt{2}) W_t^+ W_0^- = +(g^2/2) W_t^+ W_0^-.$$

The sign is **positive**: $+(g^2/2)$, confirmed by (4.5): $\text{Re}[b_t \bar{b}_0] = (g^2/4) (A_t^1 A_0^1 + A_t^2 A_0^2) \geq 0$ at $O(g^2)$. The formal identification in Landau gauge is:

$$\Delta_{\text{cross}}(t) \Big|_{\text{Landau}} / 8V \approx +(g^2/2) \cdot C_{W^\pm}(t) + O(g^3). \quad (8.3)$$

Four caveats apply:

- (i) **Perturbative validity.** Equation (8.3) holds at $O(g^2)$ and does not account for non-perturbative contributions. At $\beta_{\text{eff}} \in \{5.0, 6.0, 9.0, 12.0\}$, the $O(g^2)$ approximation is not quantitatively valid; the $O(1)$ ratio $\Delta_{\text{cross}}/G_{\text{full}} \in [0.72, 2.33]$ evidences higher-order contributions [[P4]].
- (ii) **Spatial links only.** Papers 4 and 5 use temporal gauge $A_0 = 0$, fixing temporal links to I_2 . Full Landau gauge requires temporal links; Eq. (8.3) applies to spatial links only in the current implementation.
- (iii) **Strong-coupling breakdown.** The $O(g^2)$ approximation requires lattice renormalisation to be applied systematically at finite β_{eff} .
- (iv) **Gribov copy dependence — systematic uncertainty.** The Landau gauge functional F_L generically possesses multiple local maxima on the gauge orbit (Gribov copies). Each copy gives a different value of Δ_{cross} . This systematic uncertainty is distinct from caveat (i) and is not addressed in Paper 5. This is the primary open question OQ1 (Sec. 9). See Sec. 8.4 for the proposed handling strategy.

8.4 Future Simulation Requirements

The following requirements are designated for Paper 6 and are not implemented in Paper 5:

- F5.** Steepest-descent or Fourier-accelerated gauge fixing to implement (8.2) on existing ensembles.
- F6.** Fourier-accelerated gauge fixing as an alternative for improved convergence at large volumes.
- F7.** Gribov copy strategy: ≥ 5 independent random gauge starts per configuration; select copy achieving maximum F_L ; flag configurations where F_L variation across copies exceeds 1%.
- F8.** Post-gauge-fix measurement of Δ_{cross} on gauge-fixed ensembles.
- F9.** Propagator extraction and quantitative comparison to the SU(2) Landau-gauge gluon propagator data of [[CM], [Bo]].

9 Discussion

This section synthesises the results of Sections 2–8 in prose, draws five conclusions, and identifies seven open questions. No new algebra or numerical results are introduced.

9.1 C1: Exactness and the Algebraic Status of G_{cross}

The principal achievement of this paper is the exact algebraic identification of G_{cross} . Prior to Proposition 3.1, the cross term Δ_{cross} was defined subtractionally in Paper 4 [[P4]], and its $O(1)$ magnitude ($\Delta_{\text{cross}}/G_{\text{full}} \in [0.72, 2.33]$ [[P4]]) was observed without an algebraic explanation. Theorem 3.1 provides that explanation: $G_{\text{cross}} = 8 \sum \text{Re}[b_t \bar{b}_0]$ exactly, for every $\text{SU}(2)$ configuration and every timeslice, where $b = (U)_{01}$ is the off-diagonal matrix element.

Corollary 6.1 assembles this with the a -channel faithfulness result of Proposition 1 of Paper 4 [[P4]] into a complete statement of the exact algebraic decomposition. The word “complete” in Remark 6.1 of Corollary 6.1 carries a precise meaning: together, G_{diag} and G_{cross} are sensitive to every degree of freedom in $U \in \text{SU}(2) \cong S^3$. This algebraic completeness does not imply gauge invariance, sign-definiteness, or physical observability without gauge fixing.

9.2 C2: Embedding Independence

Proposition 5.1 establishes that the rank-1 collapse $\Pi_r U \Pi_r = (U)_{rr} \Pi_r$ holds identically for all three natural tensor-product embeddings of U into $\mathbb{C}^8$ (Cases A, B, C). The spectator dimensions contribute a universal trace factor of 4, and the extracted bilinear is the same function of $(U_t)_{rr}$ regardless of embedding. The two-observable structure of $\{G_{\text{diag}}, G_{\text{cross}}\}$ is intrinsic to the fundamental spin-1/2 representation of $\text{SU}(2)$, confirmed at 1 ULP residual by checks D14 $^\pm$, D15 $^\pm$.

9.3 C3: Physical Channel Assignment

At $O(g)$ in the weak-coupling expansion, the channel assignment is exact by group-theoretic argument: T^3 is purely diagonal and contributes solely to the a -channel; $T_\pm$ are purely off-diagonal and contribute solely to the b -channel. There is no cross-contamination at this order, confirmed by checks D17-R1, D17-R2, D17-R3. The non-perturbative contributions visible in the simulated ensembles are consistent with this assignment at the algebraic level, though quantitative identification requires Landau gauge fixing and is subject to the caveats of Sec. 8.3.

9.4 C4: Gauge Properties and the Path to a Physical Observable

The gauge non-invariance of G_{cross} proved in Proposition 8.1 is expected: charged-current correlators are not gauge-invariant in pure Yang–Mills theory without gauge-

invariant composite operators or a physical gauge condition. This is a standard feature, not a pathology of the framework.

The path to a physical observable is through Landau gauge fixing (Secs. 8.2–8.4). In Landau gauge, the formal identification $\Delta_{\text{cross}}(t)|_{\text{Landau}}/8V \approx +(g^2/2) C_{W^\pm}(t)$ at $O(g^2)$ connects G_{cross} to the zero-spatial-momentum $W^\pm$ temporal correlator, with [[CM], [Bo]] as comparison targets. The positive sign $+(g^2/2)$ —confirmed by (4.5) and Sec. 8.3—means G_{cross} is non-negative at leading perturbative order; the observed negative values on thermalized configurations (37–62% of cases [[P4]]) are a genuine non-perturbative effect. Gribov copy dependence introduces a systematic uncertainty not bounded in this paper, designated OQ1.

9.5 C5: The b -Channel as Necessary Conduit; Matter for Chirality

Proposition 7.1 establishes two complementary facts. First, the b -channel is an unavoidable algebraic gateway for any gauge-link correlator sensitive to $W^\pm$: no diagonal projector correlator can access b (Lemma 5.1), so G_{cross} is the unique algebraic structure through which the charged gauge boson sector is visible. Second, the b -channel does not by itself select chirality: the pure-gauge link variable transforms identically under SU(2) regardless of matter handedness, and the distinction between left- and right-handed couplings is encoded entirely in the matter sector. The identification of G_{cross} as the algebraic conduit for $W^\pm$ is a necessary but not sufficient condition for implementing the chiral structure of the electroweak interaction. Chirality selection requires fermionic fields with definite handedness, and faces the Nielsen–Ninomiya obstruction [[NN1], [NN2]] in any local, Hermitian, doubler-free formulation.

9.6 Open Questions OQ1–OQ7

- OQ1** (*Landau gauge measurement.*) Implementing requirements F5–F9 on the existing ensembles and performing a quantitative comparison to [[CM], [Bo]]. The Gribov copy systematic (caveat (iv) of Sec. 8.3) must be characterised before the comparison can be made quantitatively reliable.
- OQ2** (*Sign-definiteness of ensemble averages.*) Whether $\langle G_{\text{diag}} \rangle$ and $\langle G_{\text{cross}} \rangle$ are sign-definite at finite β is an open numerical question requiring measurements on larger ensembles.
- OQ3** (*SU(3) extension.*) Extending the two-observable decomposition to SU(3) within the Cl(6) framework of Paper 1 [[P1]], where the algebra is larger and the projector structure more complex.
- OQ4** (*Chirality and fermionic content.*) The conditional prediction of Sec. 7.3 requires a chirally symmetric lattice fermion action. The Nielsen–Ninomiya obstruction [[NN1], [NN2]] requires domain wall or overlap fermions. Designated Future Requirements F1–F4.

OQ5 (*Electroweak mixing, $U(1)_Y$, and W/Z mass ratio.*) Including $U(1)_Y$ and the Higgs mechanism. Outside the scope of the present series.

OQ6 (*Higher-spin and reducible representations.*) Extension of the channel decomposition to higher-spin or reducible representations of $SU(2)$.

OQ7 (*Temporal gauge and full spacetime structure.*) A formulation incorporating all four-dimensional spacetime links in Landau gauge for proper continuum limit comparison and momentum-space propagator extraction.

OQ1–OQ4 are the most pressing; OQ5–OQ7 are longer-range.

10 Conclusions

We have presented a two-observable decomposition of the $SU(2)$ temporal gauge-link correlator within the $Cl(0, 2) \cong \mathbb{H}$ algebraic framework. Three results stand out as the principal contributions of this paper. We emphasise that the quaternionic parametrisation $U = a \cdot I_2 + b \cdot J$ and the resulting bilinear split are standard; the new contributions are the *projector formulation* of the channel separation, the *algebraic closure of the Paper 4 gap* (giving Δ_{cross} an independent generator), and the *Landau-gauge propagator identification* at $O(g^2)$.

Result 1 — Exact Algebraic Decomposition (Cor. 6.1; Thm. 3.1)

The pair $\{G_{\text{diag}}(t), G_{\text{cross}}(t)\}$ constitutes an exact algebraic decomposition of $G_{\text{full}}(t)$ in the $Cl(0, 2) \cong \mathbb{H}$ framework. Specifically, $G_{\text{diag}}(t) + G_{\text{cross}}(t) = G_{\text{full}}(t)$ holds exactly for every $SU(2)$ gauge configuration and every timeslice t , with $G_{\text{diag}}(t) = 8 \sum \text{Re}[a_t \bar{a}_0]$ faithful to the a -channel (Proposition 1 of Paper 4 [[P4]]) and $G_{\text{cross}}(t) = 8 \sum \text{Re}[b_t \bar{b}_0]$ faithful to the b -channel (Proposition 3.1). This resolves the structural gap of Paper 4: the cross term Δ_{cross} , previously defined subtractionally, is exactly the b -channel bilinear. Verified to $|G_{\text{diag}} + G_{\text{cross}} - G_{\text{full}}| \leq 1.78 \times 10^{-15}$ and $|G_{\text{cross}} - \Delta_{\text{cross}}| \leq 1.14 \times 10^{-13}$ across all four ensembles (D7a–D7c, D8 $\times$ 4 β).

Result 2 — Embedding Independence (Prop. 5.1)

The rank-1 collapse $\Pi_r U \Pi_r = (U)_{rr} \Pi_r$ — which forces the b -channel blindness of all diagonal projector correlators and thereby necessitates the off-diagonal structure of G_{cross} — is intrinsic to the fundamental spin-1/2 representation of $SU(2)$. It holds identically for every tensor-product embedding of U into $\mathbb{C}^8$ (Cases A, B, C of Sec. 5), with the same bilinear formula and the same spectator trace factor of 4 in each case. The two-observable structure of $\{G_{\text{diag}}, G_{\text{cross}}\}$ is not an artifact of the specific embedding $I_2 \otimes U \otimes I_2$.

Result 3 — Physical Channel Assignment and Gauge Status (Prop. 7.1; Prop. 8.1)

At $O(g)$, the generator structure of $SU(2)$ enforces an exact channel separation: $T^3 A_\mu^3$ contributes only to the a -channel and $T_\pm W_\mu^\mp$ contribute only to the b -channel (D17-R1, D17-R2, D17-R3). G_{diag} probes A^3 ; G_{cross} probes $W^\pm$. The b -channel is a necessary algebraic conduit for all charged-current coupling (Proposition 7.1). G_{cross} is proved not to be gauge-invariant (Proposition 8.1). In Landau gauge:

$$\Delta_{\text{cross}}(t) \Big|_{\text{Landau}} / 8V \approx +(g^2/2) \cdot C_{W^\pm}(t) + O(g^3),$$

with positive sign $+(g^2/2)$ confirmed by the weak-coupling calculation of Sec. 8.3.

What Paper 5 Does Not Establish

Paper 5 does not establish gauge invariance of G_{cross} or Δ_{cross} . It does not implement Landau gauge fixing, does not compute the $W^\pm$ propagator numerically, and does not perform a quantitative comparison to the lattice data of [[CM], [Bo]]. It does not address electroweak mixing, W/Z mass ratios, the Higgs mechanism, or chiral symmetry breaking. It does not implement a fermionic action of any kind and makes no claim of overcoming the Nielsen–Ninomiya obstruction [[NN1], [NN2]]. *The embedding does not, by itself, distinguish $SU(2)_L$ from $SU(2)_R$; this requires external fermionic input.* All 51 algebraic and numerical checks (D1–D18) are satisfied at tolerance $\leq 10^{-13}$ or better; the results are exact in the algebraic sense of Corollary 6.1 and Theorem 3.1, perturbative in the physical identification of Sec. 8.3, and conditional in the programme targets of Sec. 7.3.

A Algebraic Verification Checks

All 51 verification instances were executed across check IDs D1–D18. The programme achieves a complete PASS record: 51/51 instances pass, 0 fail, with 1 flagged anomaly (D10b, closed) and 3 inverted checks (D9a, D18a, D18b-n). The next available check identifier for Paper 6 is D19.

A.1 Flagged Anomaly: D10b

D10b — b -Channel Slope (Flagged; Spec Error; Closed). The log-log slope of the b -channel weak-coupling approximation error versus ε is 3.000 rather than the expected 2 specified in the original check programme. This is a specification error, not a computation error, and the check is closed with the following resolution.

The $O(\varepsilon^2)$ coefficient in the Taylor expansion of $b = (U)_{01} = (\exp(i\varepsilon H))_{01}$ vanishes identically because $H^2 \propto I_2$ for any $SU(2)$ generator H . Specifically, $(H^2)_{01} = 0$ since H^2 is proportional to the identity. The expansion of $\sin(\varepsilon \|H\|)/\|H\|$ has no ε^2 term in the off-diagonal component; the leading correction is $O(\varepsilon^3)$. This is the sin-Taylor argument. It is distinct from the BCH/ $[T_+, T_-] \propto T_3$ argument, which

explains channel purity (D17) and is a separate algebraic result. Slope = 3 confirms approximation quality; channel identity is established algebraically by Proposition 3.1, Lemma 5.1, and checks D3c–D3d.

A.2 Notes on Checks D14/D15, D16, and D18

D14/D15 — Type (b) Independent Computations. The LHS is evaluated via the full 8×8 Kronecker-product trace; the RHS accesses the 2×2 matrix element $(U_t)_{rr}$ directly. The two code paths share no intermediate data structure. The residual of $2.22 \times 10^{-16} = 1$ ULP in all four sub-checks represents genuine numerical agreement between distinct computation routes.

D16 — Consistency Check; Not Independent Verification. Check D16 evaluates the std of the diagonal projector correlator over 800 random choices of b , holding a fixed. The result $\text{std} = 0.00 \times 10^{+00}$ is guaranteed by construction: b is algebraically absent from the formula. D16 is a consistency check on the implementation. The substantive verification of b -channel blindness is established by Proposition 5.1 and the Core Lemma (5.1), and confirmed by the independent code-path comparison of D14/D15.

D18 — Scope of Gauge Non-Invariance Checks. The D18 check suite is designed for generic (Haar-random) gauge transformations. D18a measures residuals on 50 configurations, finding $\min 1.07 \times 10^{-2} \gg 10^{-6}$. D18b-f verifies (8.1) on 50 random pairs with zero floating-point residual. D18b-n verifies the specific counterexample $g_t = i\sigma_2$, $g_0 = I_2$ numerically. Near-identity transforms ($g \approx I + i\varepsilon H$, $\varepsilon \lesssim 10^{-7}$) were not included; the 10^{-6} threshold is not valid there. Gauge non-invariance under all transforms is proved analytically by Proposition 8.1.

A.3 Section-Organised Sub-Tables

Open Item ITEM-A.1. Check D18b-f was run on 5 pairs in the original execution; Appendix A reports 50 for consistency with the D5, D6, D14, D15 conventions. The result is guaranteed to remain $0.00 \times 10^{+00}$ since the verification is analytic. Extension to 50 pairs is optional and does not affect any conclusion.

B Simulation Parameters

B.1 Ensemble Parameters

The numerical results of this paper are computed on the four thermalized SU(2) pure-gauge ensembles generated in Paper 4 [[P4]]. No new Monte Carlo generation is performed in Paper 5; all new measurements are carried out on the existing configurations.

B.2 Additional Measurements on Existing Ensembles

B.3 Verification Script Summary

Open Item ITEM-B.1 (Resolved). SciPy version: 1.16.3.

Checks D1–D9 were executed within the core Paper 5 verification pipeline, building directly on the Paper 4 simulation infrastructure (`paper4_simulation.py`, `su2_corrected_T3.py`).

B.4 Paper 6 Entry Point

The simulation infrastructure of Papers 4 and 5 provides the foundation for the next stage of the programme. On the gauge side, the immediate requirement is Landau gauge fixing (F5: steepest-descent or Fourier-accelerated gauge fixing; F6: Fourier-accelerated alternative; F7: Gribov copy strategy with ≥ 5 random starts per configuration; F8: post-fix Δ_{cross} measurement; F9: propagator extraction and comparison to [\[CM\]](#), [\[Bo\]](#)).

On the fermionic side, the conditional prediction of Sec. 7.3 requires a chirally symmetric lattice fermion action (F1–F4: domain wall or overlap fermion action; construction of the cross-projected propagator $S_{\text{cross}} = P_+^{\text{LW}} S(x, y) P_-^{\text{LW}}$; measurement on gauge-fixed backgrounds).

First available identifiers for Paper 6: D19 (check); Proposition 7; Corollary 6; Theorem 2; Lemma 4.

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Table A.0: Complete check programme — D1 through D18.

Check ID	§	Description	Expected	Tol.	Result
D1 (D1a–D1d)	§1–§2	Data structure; SU(2) param; $ a ^2 + b ^2 = 1$	Param exact; (E4) holds	$\leq 10^{-15}$	PASS
D2 (D2a–D2d)	§2	Clifford grade decomp; $\text{Cl}^0 \cong \mathbb{C}$; $\text{Cl}^1 = \mathbb{C}\hat{j}$; RM1–RM5	Grade split consistent	$\leq 10^{-15}$	PASS
D3 (D3a–D3d)	§2–§3	Rank-1 collapse (diag. & off-diag.); b -blindness of P_r	Collapse formulae exact	$\leq 10^{-15}$	PASS
D4 (D4a–D4h)	§2	Generator commutation; Casimir $= (3/4) I_2$; algebra closure	Relations exact	$\leq 10^{-14}$	PASS
D5	§3	Prop. 3.1: $\text{Re Tr}[P_+^{\text{LW}} E_t P_-^{\text{LW}} E_0^\dagger] = 4 \text{Re}[b_t \bar{b}_0]$, 50 pairs	RHS matches LHS	$\leq 10^{-14}$	PASS
D6	§3	Cor. 3.1: orientation symmetry, 50 pairs	Equality holds	$\leq 10^{-14}$	PASS
D7 (D7a–D7c)	§3	Thm. 3.1: $ G_{\text{diag}} + G_{\text{cross}} - G_{\text{full}} $	$\leq 1.78 \times 10^{-15}$	$\leq 10^{-14}$	PASS
D8 ($\times 4\beta$)	§3	$G_{\text{cross}} = \Delta_{\text{cross}}$: $ G_{\text{cross}} - \Delta_{\text{cross}} $, all ensembles	$\leq 1.14 \times 10^{-13}$	$\leq 10^{-12}$	PASS
D9 (D9a)	§3	Sign non-definiteness of G_{cross} (inverted)	$G_{\text{cross}} < 0$ in ≥ 1 case	Inverted	PASS
D10 (D10a $\times 3\varepsilon$; D10b*)	§4	b -channel WC error vs ε ; log-log slope	$O(\varepsilon^3)$; slope= 3.000	$\leq 10^{-10}$	PASS*
D11 (D11a $\times 3\varepsilon$; D11b)	§4	a -channel WC error vs ε ; log-log slope	$O(\varepsilon^2)$; slope= 2.000	$\leq 10^{-10}$	PASS
D12 (D12b)	§4	Monotone $ R(\beta) $ at $t = 5$ across ensembles	Monotone in β_{eff}	Qualitat.	PASS
D13 (D13a)	§4	Cold-start $\langle b ^2 \rangle \approx 0.5$ (Haar)	≈ 0.5	$\leq 10^{-2}$	PASS
D14 (D14 $^+$, D14 $^-$)	§5	Case A ($U \otimes I_4$): 8×8 trace vs 2×2 element, $r = \pm$, 50 pairs	$\leq 2.22 \times 10^{-16}$ (1 1 ULP)		PASS
D15 (D15 $^+$, D15 $^-$)	§5	Case B ($I_4 \otimes U$): 8×8 trace vs 2×2 element, $r = \pm$, 50 pairs	$\leq 2.22 \times 10^{-16}$ (1 1 ULP)		PASS
D16	§5	b -blindness std over 800 evals	std = $0.00 \times 10^{+00}$	Exact 0	PASS
D17 (D17-R1,R2,R3)	§7	Channel purity: A^3 only in a ; $\partial(\text{Im } a)/\partial\alpha^1 = \partial(\text{Im } a)/\partial\alpha^2 = 0$	R1, R2, R3 all zero	$\leq 10^{-10}$	PASS
D18 (D18a; D18b-f; D18b-n)	§8	Gauge non-inv (inv); Lemma 8.1 formula; counterexample (inv)	$\geq 10^{-6}$; 0.00; change	Inv/Analyt.	PASS

Table A.1: Checks for Secs. 1–2 (data structure and algebraic framework).

Check ID	Description	Instances	Result
D1a–D1d	SU(2) parametrisation; $ a ^2 + b ^2 = 1$; data integrity	4	PASS
D2a–D2d	Clifford grade decomp; right $\mathbb{C}$ -module structure RM1–RM5	4	PASS
D3a–D3d	Rank-1 collapse (diagonal and off-diagonal); b -blindness	4	PASS
D4a–D4h	Generator commutation; Casimir; algebra closure	8	PASS

Table A.2: Checks for Sec. 3 (Proposition 3.1, Corollary 3.1, Theorem 3.1).

Check ID	Description	Instances	Result
D5	Prop. 3.1: off-diagonal collapse, 50 random pairs	1	PASS
D6	Cor. 3.1: orientation symmetry, 50 random pairs	1	PASS
D7a–D7c	Thm. 3.1: $ G_{\text{diag}} + G_{\text{cross}} - G_{\text{full}} \leq 1.78 \times 10^{-15}$	3	PASS
D8 $\times 4\beta$	$G_{\text{cross}} = \Delta_{\text{cross}}: \leq 1.14 \times 10^{-13}$, all ensembles	4	PASS
D9a	Sign non-definiteness of G_{cross} (inverted check)	1	PASS

Table A.3: Checks for Sec. 4 (weak-coupling expansion).

Check ID	Description	Instances	Result
D10a $\times 3\varepsilon$	b -channel weak-coupling error at 3 values of ε	3	PASS
D10b*	b -channel slope = 3 (flagged; sin-Taylor; closed)	1	PASS*
D11a $\times 3\varepsilon$	a -channel weak-coupling error at 3 values of ε	3	PASS
D11b	a -channel slope = 2	1	PASS
D12b	Monotone $ R(\beta) $ at $t = 5$	1	PASS
D13a	Cold-start $\langle b ^2 \rangle \approx 0.5$	1	PASS

Table A.4: Checks for Sec. 5 (embedding universality).

Check ID	Description	Instances	Result
D14 ⁺ , D14 [−]	Case A ($U \otimes I_4$): type (b) independent, $r = \pm$, 50 pairs	2	PASS
D15 ⁺ , D15 [−]	Case B ($I_4 \otimes U$): type (b) independent, $r = \pm$, 50 pairs	2	PASS
D16	b -blindness std = $0.00 \times 10^{+00}$, 800 evals (consistency check)	1	PASS

Table A.5: Checks for Sec. 7 (channel purity).

Check ID	Description	Instances	Result
D17-R1	$\text{Im}(a_\varepsilon - 1)/\varepsilon = \alpha^3/2$; $T^3 A^3$ only	1	PASS
D17-R2	$\partial(\text{Im } a)/\partial\alpha^1 = 0$; no A^1 contamination	1	PASS
D17-R3	$\partial(\text{Im } a)/\partial\alpha^2 = 0$; no A^2 contamination	1	PASS

Table A.6: Checks for Sec. 8 (gauge non-invariance).

Check ID	Description	Instances	Result
D18a	Gauge non-inv, statistical; min 1.07×10^{-2} ; 50 configs (inverted)	1	PASS
D18b-f	Lemma 8.1 formula (8.1): analytic, 50 pairs; residual $0.00 \times 10^{+00}$	1	PASS
D18b-n	Counterexample $g_t = i\sigma_2$, $g_0 = I_2$: $b'_t = a_t^*$ detected (inverted)	1	PASS

Table B.1: Ensemble parameters.

Parameter	Values / Setting
Effective coupling β_{eff}	{5.0, 6.0, 9.0, 12.0}
Spatial extent N	6 (i.e., 6^3 spatial lattice)
Configurations per ensemble N_{cfg}	20
Thermalisation sweeps N_{therm}	500
Metropolis acceptance rate	$\approx 45\%$
Gauge condition	Temporal gauge ($A_0 = 0$; temporal links fixed to I_2)
Random seed	42
Initialisation	Cold start ($U_\mu(x) = I_2$ for all links)

Table B.2: New measurements on Paper 4 ensembles.

Measurement	Description	Check IDs
M1	$G_{\text{diag}}(t)$, $G_{\text{cross}}(t)$ per configuration per timeslice t	D7, D8
M2	$\Delta_{\text{cross}}(t)$ comparison: $ G_{\text{cross}} - \Delta_{\text{cross}} $ per ensemble	$D8 \times 4\beta$
M3	Compact identity: $ G_{\text{diag}} + G_{\text{cross}} - G_{\text{full}} \leq 1.78 \times 10^{-15}$	D7a–D7c
M4	Sign diagnostics: fraction of (cfg, t) pairs with $G_{\text{cross}} < 0$	D9a
M5	Gauge transformation test: Δ_{cross} before and after Haar-random $g(x)$	D18a

Table B.3: Verification script assignments.

Script	Checks	Sections
paper5_verification_s4.py	D10a, D10b, D11a, D11b, D12b, D13a	§4
paper5_verification_s5.py	D14 ⁺ , D14 [−] , D15 ⁺ , D15 [−] , D16	§5
paper5_verification_s7.py	D17-R1, D17-R2, D17-R3	§7
paper5_verification_s8.py	D18a, D18b-f, D18b-n	§8